Torrent 12FFC Multi-Protocol PHY

TSMC 12nm FFC

Part #: SD0801

IP Rev: R130 - Doc Rev: 1.2

Quick Start Guide

September 02, 2020

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1 Introduction

This document is intended for developers working with the Torrent 12FFC Multi-Protocol PHY IP. This document provides the user with information about the IP delivery package, including its structure, naming, and included content. It also provides a description of the specifications and guides in the delivery (see *Section 5 Documentation* on page 16). References to specific sections of these documents are provided in this document in an effort to quickly guide customer teams to information of interest.

1.1 Release Information

Table 1 lists the release information of Torrent 12FFC Multi-Protocol PHY IP.

Item	Description
IP Name	Torrent 12FFC Multi-Protocol PHY
Version	R130
Product Part Number	SD0801

Table 1: Release Information

1.2 What's in the Release

- Two IP directories, one for the hard macro (PMA), and one for the soft PHY wrapper (PHY)
- A RELEASE_TREE file detailing all directories and file contents in the delivery package.
- A RELEASE_NOTES file containing basic information about the current release version.

1.3 Hard Macro Naming Convention Details

Hard macro product and directory names contain information about several configuration parameters. The name format is:

cdn_<ProductID>_<TechnologyNode>_<LaneConfiguration>_<Package/Bump>_<MetalStack>

Fields in the macro name are defined as follows:

ProductID	This is the unique product ID that is used to identify all the different implementations of this PHY product. For this product an ID of ***sd0801*** is used to delineate a standard multi-link multi-protocol PHY.
TechnologyNode	This field specifies the technology process node used for hard macro implementation. For this specific macro the field value is ***t12ffc***, representing TSMC 12FFC technology.
LaneConfiguration	This field specifies the lane configuration of the macro. This is a two-digit field in which the first digit specifies the number of lanes on the left side of the common module and the second digit specifies the number of lanes on the right side of the common module.
Package/Bump	This field is a 5-character string that represents the packaging and bump pitch details for the hard macro implementation. At present only a value of ***vf162*** indicating standard flip-chip packaging with 162 micron bump pitch, is supported for this IP.
MetalStack	This field specifies the technology metal stack used in the hard macro implementation. The following are the currently supported values for this field: ***2xa1xdh3xevhv2z, 2xa1xdh3xevhv2y2r, 2xa1xdh4xevhvh2r.***

1.4 Soft PHY Naming Convention Details

Soft PHY product and directory names carry generally the same format as that described for the hard macro in the previous section. This approach allows both Cadence support teams and customers to quickly identify the type of macro that is integrated into the soft PHY wrapper. Since PHY wrappers also have many different configurations, however, the PHY naming adds a string suffix to the Product ID field to provide information about the features included in the wrapper. This suffix will take one of the following three forms:

1. PIPE-Only Interface – Suffix format ***p<b/s><w/n><a/x>***, where:
   a. “p” serves as a PIPE-only interface identifier
   b. “b/s” indicates basic (b) or source-synchronous (s) PIPE interface
   c. “w/n” indicates wide (w) 32-bit or narrow (n) 16-bit internal PIPE PCS datapath
   d. “a/x” indicates automotive (a) or non-automotive (x) application support
2. Raw SerDes Interface – Suffix format ***r<f/c><e/n><a/x>***, where:
   a. “r” serves as a raw SerDes interface identifier
   b. “f/c” indicates a full (f) or constrained/reduced (c) raw interface to the PMA hard macro
   c. “e/n” indicates 10G-KR Ethernet (e) support or no presence of such support (n)
   d. “a/x” indicates automotive (a) or non-automotive (x) application support
3. PIPE + Raw SerDes Interface – Suffix format ***m<b/s><w/n><f/c><e/n><a/x>***, where:
   a. “m” serves as a multi-purpose PIPE + raw SerDes interface identifier
   b. “b/s” indicates basic (b) or source-synchronous (s) PIPE interface
   c. “w/n” indicates wide (w) 32-bit or narrow (n) 16-bit internal PIPE PCS datapath
   d. “f/c” indicates a full (f) or constrained/reduced (c) raw interface to the PMA hard macro
   e. “e/n” indicates 10G-KR Ethernet (e) support or no presence of such support (n)
   f. “a/x” indicates automotive (a) or non-automotive (x) application support

1.5 Delivery Package Naming Convention Details

Since the PHY soft wrapper is considered the top level of the multi-protocol PHY IP, the top level delivery package naming will follow the soft PHY naming convention with an additional revision suffix added. This suffix allows both customers and Cadence support teams to distinguish between different IP releases and will take the form ***rXvYpZ***, where:

1. “rX” indicates the major product revision number for a given IP. Revision numbers (represented by “X”) begin at 100 and are incremented by 100 if major functional changes are made, by 10 if small internal revisions or bug fixes are made, and by 1 if metal-only changes are made.

2. “vYpZ” indicates sub-versions for each IP revision. Initial revisions will have sub-versions of v0p1, while final production versions will have sub-versions of v1p0. Other supported interim sub-versions of v0p3, v0p5, v0p7, and v0p9 are also possible depending on the relative stability and completion level of the included deliverables. If subsequent releases of a given major and minor IP version are made, an additional alphabetic character (e.g. v0p7a, v0p7b, etc) will be added to the end of this field to enable customers and Cadence IP support to differentiate between the releases.

2 Delivery Structure

This section defines the directory structure used in customer deliveries of Cadence hard and soft IP products. The intent is to define the individual subdirectories required for each type of IP delivery, even if the directory contents are not provided in a given delivery.

2.1 Top Level Directory Structure

The top level directory of each customer delivery is defined as an encapsulating directory containing subdirectories for each type of IP in the release. Individual hard and soft IP design components reside in their own specific subdirectories, one subdirectory per component, and are appropriately named based on conventions described in*Section 1 Introduction* on page 6. Contents of the top level directory are detailed in *Table 2*.

Directory/File	Description
RELEASE_NOTES.pdf	File providing release details for delivery (RTL tags used, simulation instructions, design status, etc)
RELEASE_TREE	File containing output of Linux “tree” command, providing hierarchical listing of every directory and file in the delivery
<hard_IP_name>	Directory containing all delivery collateral related to a given hardened macro. Each macro is contained in its own individual directory.
<soft_IP_name>	Directory containing all collateral related to a given soft IP delivery. Each soft unit is contained in its own individual directory.

Table 2: Top Level Delivery Directory Contents

The top level structure is graphically depicted below in *Figure 1*. Note that a given delivery package may contain multiple hard and/or soft IP block subdirectories in addition to the supporting release note and tree files.

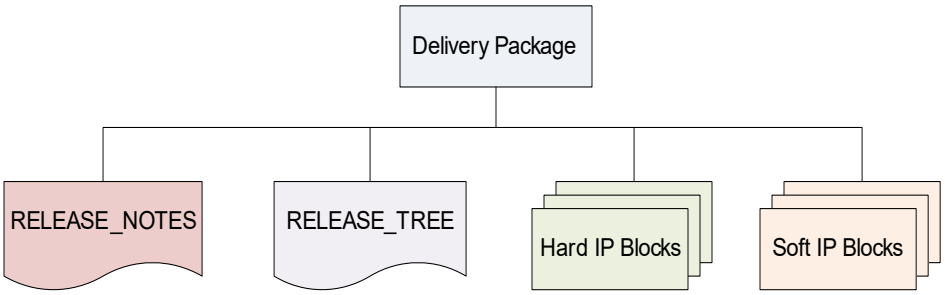

Figure 1: IP Delivery Package Structure

2.2 Design Delivery Subdirectories

Customer delivery subdirectory contents are based on a set of directory definitions intended to accommodate each possible type of design view or data set that could generally apply to Cadence IP. Hard and soft IP deliveries each use a unique subset of these directories. This means that hard and soft IP deliveries from all Cadence groups will have a consistent look and feel. The superset of delivery directories, along with brief content definitions, is provided in *Table 3*.

Directory/File	Description
dft	Encapsulating directory containing comprehensive information regarding DFT support for a given IP block. Includes subdirectories and collateral related to both internal scan and boundary scan functionality as necessary. Information includes scan abstracts, boundary scan chain descriptions in BSDL format, and ATPG vector generation and simulation guidelines.
doc	Directory containing all relevant IP documentation. Typical contents include design specifications, integration guides, user guides, testbench descriptions, and high-volume manufacturing information.
func_ver	Encapsulating directory containing a variety of different collateral sets related to functional verification of a given IP block. This directory contains the following subdirectories: • gatesim Gate simulation support information • hvm High-volume manufacturing test environment • sanity Demonstration sanity-level test environment • soma Cadence VIP and memory model SOMA files • sva System Verilog assertions • sdf Delay back-annotation files
fpga	Directory containing scripts and other information related to FPGA implementations incorporating or interfacing with a given IP block.
gds	Directory containing physical GDS-II representation of a hardened IP block.
models	Encapsulating directory containing various models/views of a given IP block. This directory contains the following subdirectories, each representing a different design model: • cdb Celtic database noise models • cmm Apache power models • curr_prof Data related to circuit current profiles • db Compiled versions of hardened macro Liberty timing abstracts • ibis BIS I/O models • ipxact Register abstraction models in IP-XACT format • lef Hardened macro physical abstract • lib Hardened macro Liberty timing abstracts • rdl Register description language abstracts • sparam S-parameter files for channel de-embedding • udn User-defined noise models
netlists	Encapsulating directory containing various netlist formats for a given IP block. This directory will generally be used by hard IP, with any soft IP netlists generated by trial or demonstration synthesis operations included in the *synth* subdirectory. This directory contains the following subdirectories, each representing a different netlist type: • cdl Transistor-level netlist • def Design Exchange Format layout netlist • spectre Spectre simulation netlist • spice Spice simulation netlist • verilog_gate Verilog gate-level netlists

Table 3: Design Delivery Subdirectories

Directory/File	Description
pwr_intent	Encapsulating directory containing files related to the description of the power intent of a given IP block. Aside from the power format files, the subdirectories of this directory may also contain supporting collateral such as power-aware Liberty models of internal blocks and other information related to power format usage. This directory contains the following subdirectories: • cpf Common power format collateral • upf Universal power format collateral
phys_des	Directory containing information related to physical implementation details and consideration for a given IP block. Usage and definition of this directory and its contents are still being developed.
phys_ver	Encapsulating directory containing physical verification results from PVS and Calibre. Content for the Calibre subdirectory is likely to vary from project to project (if included at all), but it is currently expected that, at minimum, the following will be present in the PVS subdirectory: • ant Results of antenna checks • cmp Chemical-mechanical polishing check results • drc Results of IP-level DRC checks • lpa Litho physical analysis check results • lvs Results of layout versus schematic checks • pkg_cbp Copper bump pillar package DRC results • pkg_fc Flip-chip package DRC results • pkg_wb Wirebond package DRC results • xor XOR results (mainly in metal-only updates) • rto RTO results (mainly in metal-only updates)
software	Directory containing software drivers and any other software-related support collateral for a given IP block. Content and definition of subdirectories in this directory, if any, is still under development.
hdl_qc	Directory containing various delivery collateral related to HDL quality control analysis results (lint, CDC, etc). Full definition of content to be delivered to customers is still being developed, but it is generally expected that certain items like lint and CDC waiver sets will be included here.
sta	Encapsulating directory containing data, files, and information related to static timing analysis support for a given IP block. This directory contains the following subdirectories (the exception being that SPEF files will not be delivered for soft IP): • constraints Timing constraints, perhaps in varied formats • logs Log files of trial timing runs • reports Analysis reports based on provided constraints • technology Technology-specific setup files • scripts Support scripts for STA run and analysis • spef Extracted parasitic files for netlist annotation
synth	Encapsulating directory containing data, files, and information related to logic synthesis for a given IP block. This directory contains the following subdirectories: • constraints Functional and scan mode constraint files • logs Log files of trial synthesis runs • reports Analysis reports based on provided constraints • technology Technology-specific setup files • scripts Support scripts for synthesis run and analysis

Table 3: Design Delivery Subdirectories (Continued)

Directory/File	Description
hdl	Encapsulating directory for design source files. This directory contains the following subdirectories: • behav_enc Encrypted behavioral models • behav_src Unencrypted behavioral models • hdl_enc Encrypted synthesizable design source files • hdl_src Unencrypted synthesizable design source files

Table 3: Design Delivery Subdirectories (Continued)

2.3 Soft IP Delivery Directory Structure

The delivery directory structure for soft IP blocks, extracted from the superset list of directories defined in *Table 3*, is depicted below in *Figure 2*. Colors in the diagram are intended only to make it easier to identify primary directories and the subdirectories that accompany them – no further relationship between content of similarly-shaded directories should be inferred. All soft IP deliveries originating from Cadence will follow this structure, even if certain directories are not populated for a given IP block.

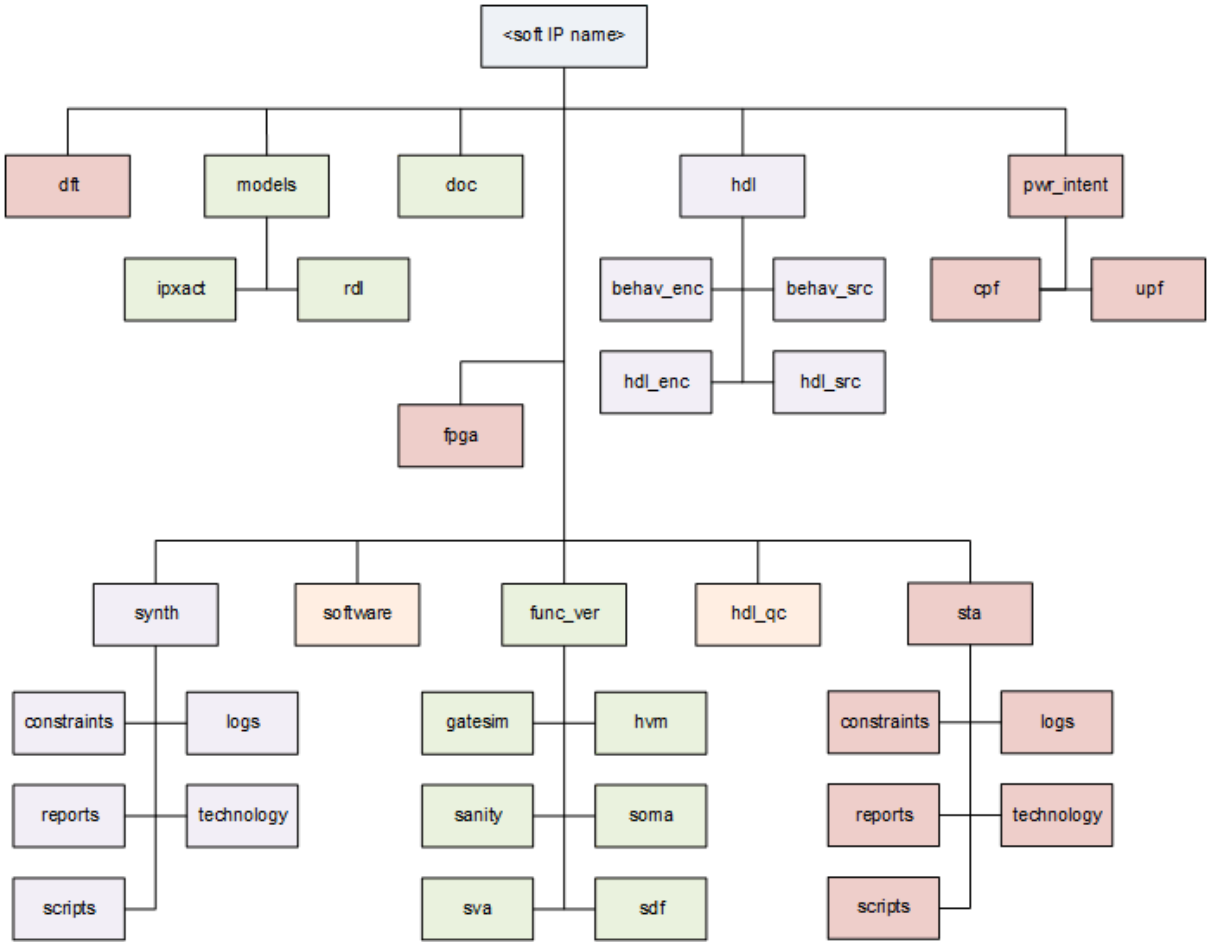

Figure 2: Soft IP Delivery Directory

2.4 Hard IP Delivery Directory Structure

The delivery directory structure for all IPG hard IP blocks, extracted from the superset list of directories defined in *Table 3*, is depicted below in *Figure 3*. As with the soft IP structure diagram shown previously, colors in the diagram are intended only to make it easier to identify primary directories and the subdirectories that accompany them. All hard IP deliveries originating from Cadence will follow this structure, even if certain directories are not populated for a given IP block.

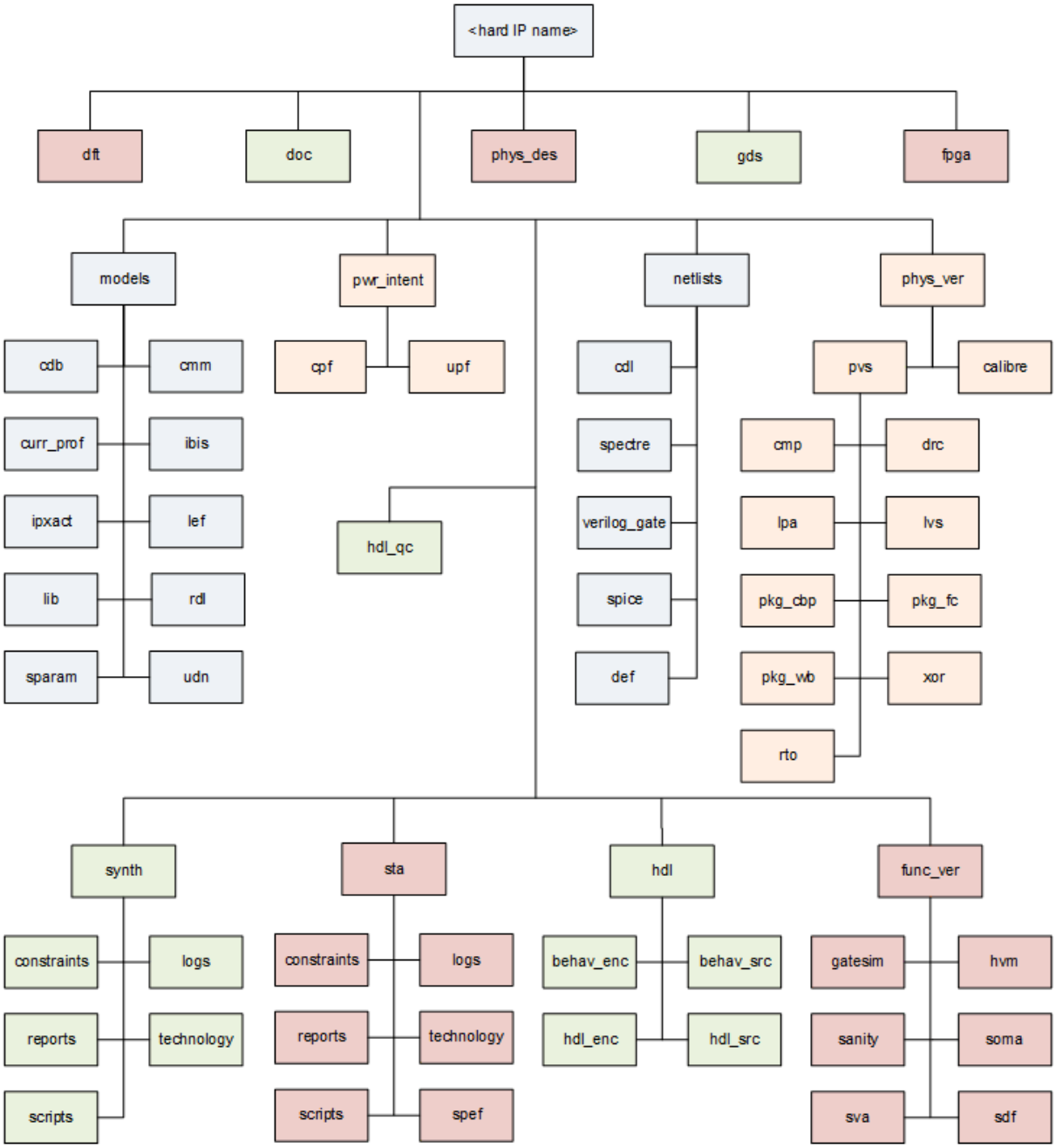

Figure 3: Hard IP Delivery Directory

3 Setting Up the Environment

3.1 Install IP Delivery Package

The complete IP delivery package is delivered as a single tarred/gzipped file, named based on conventions described in *Section 1.5 Delivery Package Naming Convention Details* on page 7, that includes all necessary file and directory collateral for a given IP revision. As noted previously, once installed the top IP directory contains the RELEASE_NOTES and RELEASE_TREE files as well as two IP directories for the subcomponents that comprise the entire PHY unit (soft PHY and hard PMA).

Place the delivery tarball in your desired location (subsequently referred to as ***IP_INSTALL_PATH***) and unpack the contents using the standard Unix/Linux command:

gtar xvfhz <IP_DELIVERY>.tgz

After the unpacking process is completed it is recommended that a directory tree listing be generated and compared to the included RELEASE_TREE file to insure all the expected delivery collateral is present.

3.2 Define RTL Base Path Variable

Both hard and soft IP delivery packages make use of an ***RTL_HOME*** environment variable to set the base path for various scripts and manifest files. This variable should be set to the full path of the top level IP package directory, after unpacking and installation, using the standard Unix/Linux ***setenv*** command:

setenv RTL_HOME <IP INSTALL PATH>/<IP DELIVERY>

3.3 Define Standard Cell Library Base Path Variable

Full production hard IP delivery packages contain gate-level netlists to support functional gate simulation as well as DFT vector generation and simulation. For ease of integration a standard cell library manifest file has been provided that specifies the technology libraries and versions needed to compile these netlists. Since customer library installation paths will differ from those used by Cadence development teams this manifest file includes an environment variable that must be set to identify the base installation path to the standard cell Verilog models. Customers should examine the library information in the ***<IP_INSTALL_PATH>/<IP_DELIVERY>/<HARDIP>/netlists/verilog_gate/<HARDIP>_stdcells.f*** file and set the provided ***CDN_SD0801_T12FFC_STDCELL_VLOG_HOME*** variable as necessary using the standard Unix/Linux setenv command:

setenv CDN_SD0801_T12FFC_STDCELL_VLOG_HOME <BASE_TECH_LIBRARY_PATH>

4 Design Compilation and Demo Simulation

Each soft IP delivery includes a basic sanity-level demonstration testbench that allows customers to (a) compile the complete design to insure proper installation and setup and (b) see examples of proper IP startup and basic data transfers. When practical (i.e. when design is of sufficient maturity) the testbench collateral will also provide infrastructure for compiling gate netlists for functional simulation and annotating SDF delay data onto relevant portions of the design. Information on compilation and simulation support is provided in the following subsections.

The provided scripts are based on Cadence tools and their associated command lines, and it is generally expected that customers who use Cadence simulators should be able to execute these scripts without modification. Customers using other simulators can still use the existing scripts to understand the various files and switches needed to enable the runs.

4.1 Prerequisites

Successful compilation and simulation requires that the IP has been properly unpacked and installed and that necessary environment variables have been set as described in *Section 3 Setting Up the Environment* on page 13.

4.2 Basic Compilation

1. Make sure that prerequisites are met as noted above.

2. Make a run directory in any desired work location as scripts are designed to operate regardless of run location

   mkdir <work_area_dir>

3. Change to the newly created directory

   cd <work_area_dir>

4. Run the provided compile script

   %unix> <IP_INSTALL_PATH>/<IP_DELIVERY>/<SOFTIP>/func_ver/sanity/scripts/compile.sh

5. Review the resultant log file. Logs will have a *PASS/FAIL/INCOMPLETE* suffix to reflect the outcome of the compilation exercise.

4.3 Basic Simulation

1. Make sure that all prerequisites are met as noted above

2. Make a run directory in any desired work location as scripts are designed to operate regardless of run location

   mkdir <work_area_dir>

3. Change to the newly created directory

   cd <work_area_dir>

4. Run the provided simulation script, providing a specific test name as the argument. Supported tests can be found in the ***<IP_INSTALL_PATH>/<IP_DELIVERY>/<SOFTIP>/func_ver/sanity/tests*** directory. By default, basic simulation execution as shown here will generate a waveform database, compile the design without power/ground pins, and run without speeding up the IP startup procedure.

   %unix> <IP_INSTALL_PATH>/<IP_DELIVERY>/<SOFTIP>/func_ver/sanity/scripts/runsim.sh <test_name>

5. Review the resultant log file. Logs will have a base file name matching the test name and will include a *PASS/FAIL/INCOMPLETE* suffix to reflect the outcome of the compilation exercise.

4.4 Running Advanced Sanity Simulation Scripts

As noted in *Section 4.3 Basic Simulation* on page 14 the default operation of the *runsim.sh* script is to execute a basic simulation which does not compile power/ground pins into the design, runs without speeding up the IP startup procedure, and creates a waveform database of simulation results. Advanced command line options are supported to modify this behavior, as follows:

1. Waveform dumps can be disabled by adding ***no_waves*** to the command line

2. IP startup can be shortened by adding ***speedup*** to the command line

3. Power/ground pins can be added to the RTL compilation process by adding ***pg_pins*** to the command line

5 Documentation

The following documents are provided as part of the full set of customer deliverables for the Torrent 12FFC Multi-Protocol PHY IP to support customer enablement in integrating and using the IP in an SoC design.

Cadence Torrent 12FFC Multi-Protocol PHY Product Datasheet - The product datasheet provides high level information on Torrent 12FFC Multi-Protocol PHY architectural features, benefits, and functional operations of the IP. The target audience for this product datasheet are the architects and procurement / engineering management looking to learn about Cadence IP.

Cadence Torrent 12FFC Multi-Protocol PHY Specification - This specification describes the functional operations of the Torrent 12FFC Multi-Protocol PHY, an implementation of a Multi-protocol PCIe/USB/SATA/Ethernet PHY, consisting of a PIPE compliant multi-protocol (PCIe Gen1/2/3, USB 3.0/3.1) PCS, IEEE 802.3 clause 72 link training modules, supporting components (TAP controller, APB interface, PHY power control, clock/reset controller) and PMA hard macro.

Cadence Torrent 12FFC Multi-Protocol PHY- PMA Specification - This specification describes in detail the functional operations of the Torrent 12FFC Multi-Protocol PHY hard macro, an implementation of a multi-protocol physical media attachment sub-layer (PMA). It is intended for those involved in using the macro in a system-on-chip environment.

Cadence Torrent 12FFC Multi-Protocol PHY Integration Guide - The integration guide describes the strategy for integrating the Torrent 12FFC Multi-Protocol PHY in a System-on-a-Chip (SoC) environment. It is intended for those involved in the process of product design integration at both logical and physical layers. The integration guide provides specific hard macro integration instructions and guidelines, including those for simulation, synthesis, and production test strategies to facilitate customer enablement. It also provides mission-mode and test feature integration information for both hard and soft IP layers.

Cadence Torrent 12FFC Multi-Protocol PHY User Guide - The user guide describes the architectural features, hardware functionality and supported configurability of the Torrent 12FFC Multi-Protocol PHY IP design. It is intended for those involved in the process of product design setup, configuration and verification.

Cadence Torrent 12FFC Multi-Protocol PHY Programmer's Guide - The programmer's guide provides specific steps on how to configure the Torrent 12FFC Multi-Protocol PHY to enable operation in each of the supported functional modes. Specific register details are provided for each protocol and clocking option, and step-by-step procedures are defined for many common PHY operations. It is intended for those involved in development of software/firmware to support PHY operation in the end-use system and for those performing verification of the logical SoC functions.

5.1 Work-flow – Documentation Map

Figure 4 shows a task-based work-flow chart that outlines common steps performed in the process of consuming third-party IP and integrating the IP into a larger system. The intent of the chart is to provide a reference point to allow teams responsible for each step of this process to quickly access relevant information included in the Cadence-supplied documentation set.

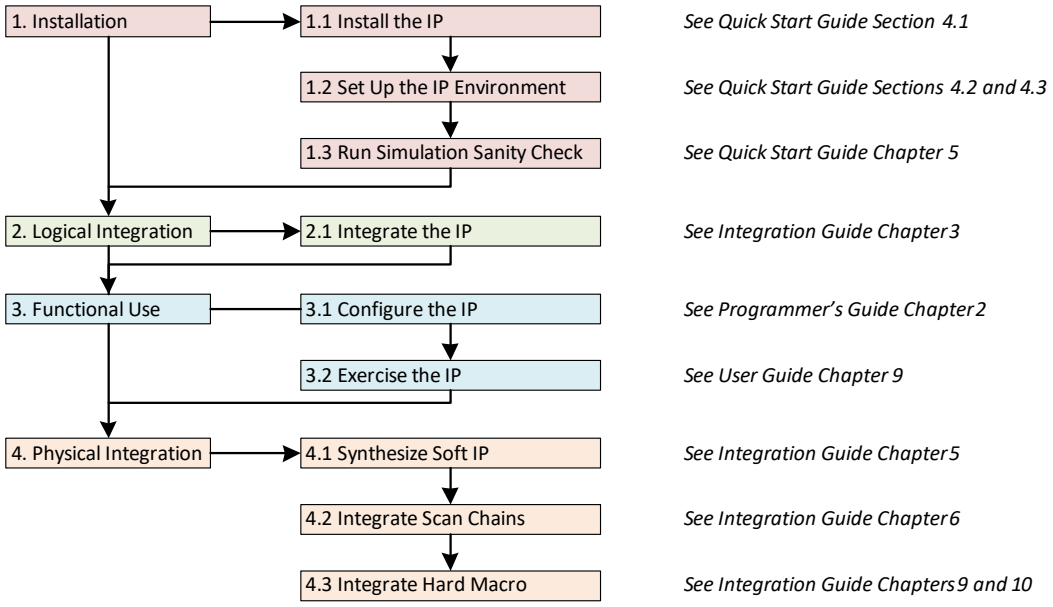

Figure 4: Work-flow – Documentation Map

Change Log

Version	Description of changes
1.1	Release Date: February 19, 2019 Created Torrent 12FFC R120 version of quick start guide
1.2	Release Date: September 02, 2020 Created Torrent 12FFC R130 version of quick start guide